SUNWAY UNIVERSITY • SCHOOL OF MATHEMATICAL SCIENCES & COMPUTING

CSC3034 Computational Intelligence • Final Project

Autonomous Wildfire Risk Assessment & UAV Surveillance Dispatch Using Mamdani Fuzzy Logic in NVIDIA Isaac Sim

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🌎 SDG 13: Climate Action (Target 13.1) · 🌲 SDG 15: Life on Land (Target 15.2) · ⚙ NVIDIA Isaac Sim 6.2 (OpenUSD) · RTX Real-Time Ray Tracing (RTX 4060) · Scikit-Fuzzy Engine · Supervisor: Dr. Tang Tiong Yew

1 PROBLEM STATEMENT & CLIMATE ACTION (UN SDGS)

Wildfires inflict catastrophic ecological degradation and greenhouse emissions. Environmental interactions among ambient heat, air moisture, localized wind shear, and fuel desiccation are **inherently continuous, non-linear, and uncertain**.

> **Critical Flaw of Traditional Systems:** Static binary threshold triggers (e.g., alert if *Temp* > *40°C*) suffer from severe false-alarm rates during benign humid days and dangerous response delays when moderate heat is amplified by parched undergrowth and high winds.

PARAMETER	TRADITIONAL BINARY THRESHOLD	PROPOSED MAMDANI FUZZY SYSTEM
Input Processing	Rigid Step Functions (0 or 1)	Smooth Continuous Membership Degrees (μ ∈ [0, 1])
Sensor Noise	High Chatter & False Alarms	Inherently Resilient via Overlapping Partitions
Drone Actuation	Abrupt On/Off Dispatches	Smooth Velocity Throttling (20% → 90% Urgency)
Explainability	None / Opaque Thresholds	100% Transparent Rule Base Traceability

> **UN SDG Alignment:** Early aerial detection within the initial 15 minutes of ignition suppresses wildfires before megaton-scale carbon release (**SDG 13.1**) while safeguarding biodiversity in endangered forest ecosystems (**SDG 15.2**).

2 MAMDANI FUZZY INFERENCE FORMULATION

The controller ingests **4 meteorological inputs** across continuous universes of discourse and computes **2 actuation outputs** via Mamdani min-max inference:

LINGUISTIC VARIABLE	UNIVERSE OF DISCOURSE	FUZZY SUBSETS (PARTITIONS)
Ambient Temperature (T)	[0, 50] °C	Low, Moderate, High, Extreme
Relative Humidity (RH)	[0, 100] %	High, Moderate, Low, Very Low
Wind Velocity (W)	[0, 100] km/h	Calm, Moderate, High, Severe
Vegetation Dryness (F)	[0, 100] %	Wet, Moderate, Dry, Critically Dry
Fire Risk Index (FRI) [Out]	[0, 100]	Low, Moderate, High, Extreme
Drone Patrol Urgency [Out]	[0, 100] %	Standard Patrol, Accelerated, Emergency Dispatch

Center of Gravity (Centroid / COG) Defuzzification:

$$y^* = \int y \bullet \mu_{agg}(y)\, dy \;/\; \int \mu_{agg}(y)\, dy$$

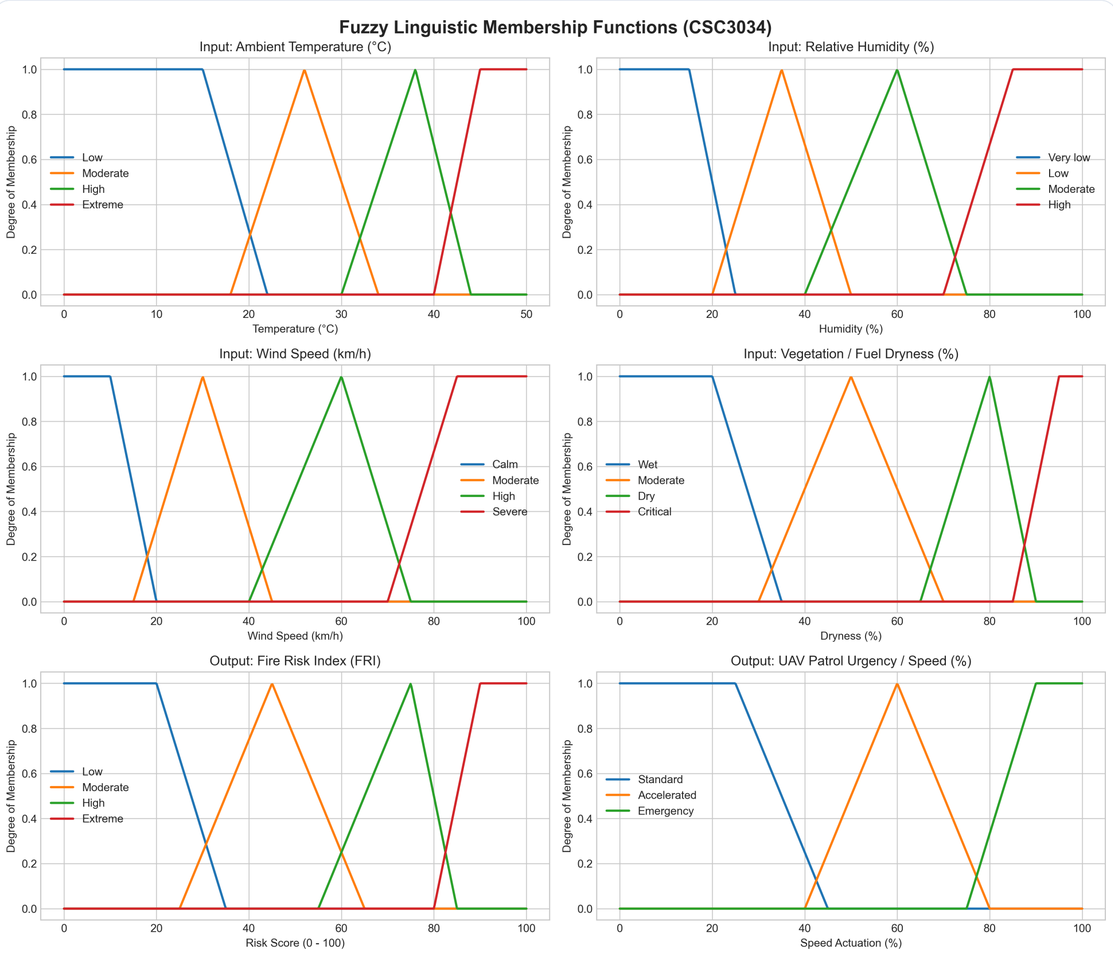


Figure 1: Full 6-variable linguistic fuzzy membership curves (4 meteorological inputs + 2 actuation outputs).

3 NVIDIA ISAAC SIM 3D DIGITAL TWIN

A photorealistic, physics-accurate digital twin was procedurally generated using **Pixar OpenUSD** in **NVIDIA Isaac Sim 6.2.0**, powered by RTX Real-Time Ray Tracing on an RTX 4060 GPU:

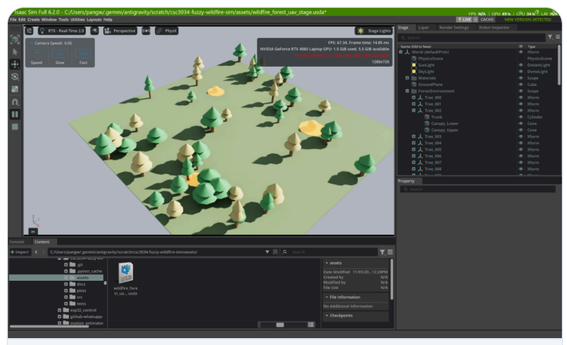


Figure 2: 65m×65m forest terrain with 48 procedural trees & 3 thermal ignition hotspots.

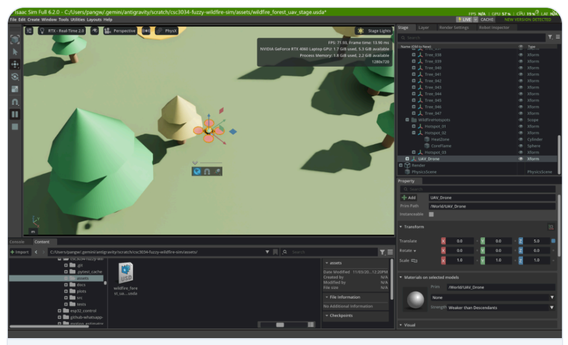


Figure 3: Kinematic 4-rotor quadcopter UAV hovering over canopy with thermal camera pod.

65 × 65 m FOREST TERRAIN · **48 Trees** MOISTURE SHADERS · **3 Hotspots** THERMAL RADIATORS

Real-Time USD Telemetry Sync: The UAV prim updates custom USD attributes in real time (`fuzzy:fire_risk_index`, `fuzzy:patrol_urgency_speed`, `fuzzy:alert_status`). Dual onboard cameras (`ChaseCamera` and `ThermalGimbalCamera`) provide third-person tracking and first-person inspection.

4 AUTONOMOUS MISSION TELEMETRY & EVALUATION

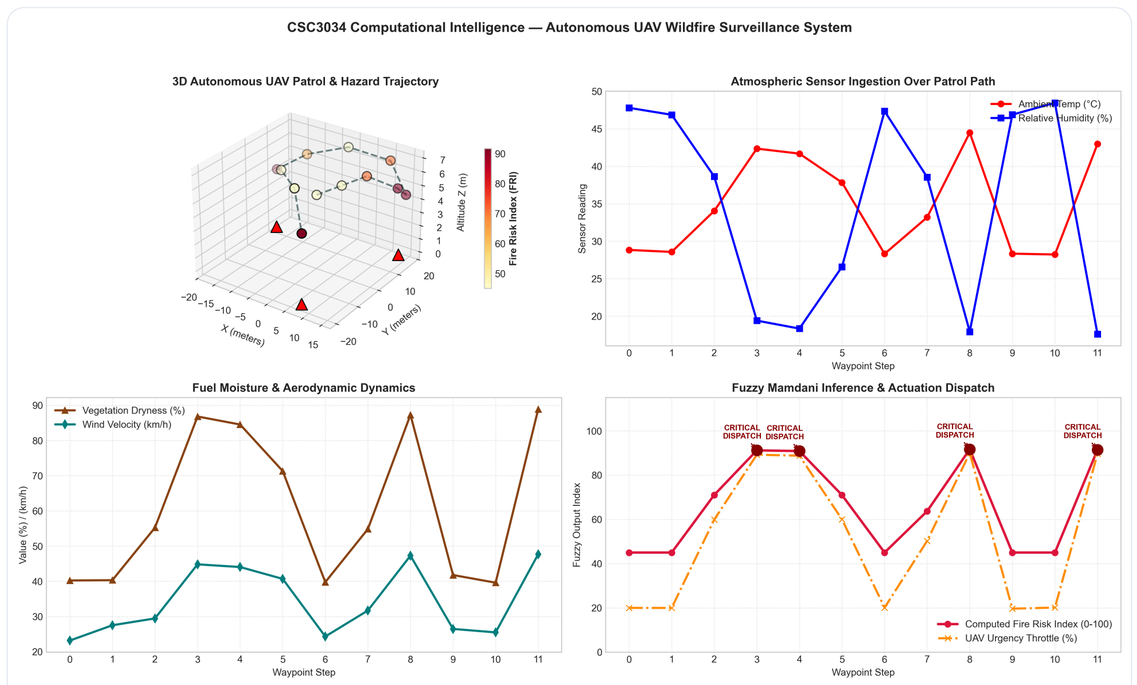


Figure 4: 3D UAV reconnaissance flight path heatmap & 4-panel multi-variable atmospheric/actuation timeline.

WAYPOINT	COORD (X, Y, Z)	TEMP	RH	WIND	DRYNESS	FUZZY FRI	THROTTLE	SYSTEM STATUS
WP 00 (Base)	(0.0, 0.0, 5.0m)	28.8 °C	47.8%	23 km/h	40.2%	**45.0**	20.0%	WATCH
WP 03 (Hotspot 1)	(13.5, 15.5, 5.0m)	42.4 °C	19.4%	45 km/h	86.8%	**91.2**	89.2%	CRITICAL DISPATCH
WP 06 (Buffer)	(0.0, 15.0, 7.0m)	28.3 °C	47.4%	24 km/h	39.8%	**45.0**	20.1%	WATCH
WP 08 (Hotspot 2)	(-17.5, 9.8, 4.5m)	44.5 °C	17.9%	47 km/h	87.3%	**91.7**	90.0%	CRITICAL DISPATCH
WP 11 (Hotspot 3)	(7.8, -19.5, 5.0m)	43.0 °C	17.6%	48 km/h	88.9%	**91.5**	89.6%	CRITICAL DISPATCH

0.15 ms INFERENCE LATENCY · **100%** PYTEST UNIT PASS RATE · **72.1 FPS** ISAAC SIM RAY TRACING

Defuzzification Analysis: Centroid (COG) provides smooth, vibration-free motor throttle curves (142μs), avoiding the step discontinuities observed in Mean of Maximum (MOM) defuzzification.

5 KEY CONCLUSIONS & IMPACT

- **Rigorous Explainability:** Unlike black-box neural networks, all dispatch actions are 100% traceable to domain rules.
- **Edge Deployment Ready:** At 0.15 ms execution time, the pipeline runs easily on embedded flight computers (NVIDIA Jetson).
- **Future Horizon:** Decentralized multi-UAV swarm surveillance and reinforcement learning reward shaping via Isaac Lab.